SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.5: Moments of Equilibrium — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.5: Moments of Equilibrium
Driving Question	DRIVING QUESTION: Why do some objects stay balanced while others topple — and how can we use the physics of turning forces to predict and control whether an object will be stable or fall?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Why Did the Stool Fall? — Introducing the Moment of a Force**	Students observe a three-legged jiko stool being nudged off-centre and toppling, then compare it with a wide-based mkebe (tin) that remains upright under the same nudge. Teacher should allow 30 seconds of silent observation before cold-calling 3 students: 'What was different about the two objects just before one fell?' Record all student predictions on the Driving Question Board (DQB) without correcting them at this stage.	A force acting at a distance from a fixed point (pivot) produces a MOMENT — a turning effect. The moment of a force is calculated using the formula $M = F \times d$, where F is the applied force in Newtons (N) and d is the perpendicular distance from the pivot to the line of action of the force, measured in metres (m). The SI unit of moment is the Newton-metre (N·m). A larger force or a greater distance from the pivot produces a larger turning effect. Pedagogical note: Stress 'perpendicular distance' carefully — students frequently measure along the object rather than perpendicularly to the force's line of action.	The stool toppled because the turning effect (moment) created by its displaced weight about the base edge exceeded the restoring moment. Using $M = F \times d$: if a 5 N sideways push is applied 0.4 m from the pivot edge, the moment = 5 × 0.4 = 2.0 N·m. Teacher should work through two Kenyan-context numerical examples: (1) a Kericho tea-picker's basket hung 0.3 m from a shoulder pivot under 12 N load; (2) a Nairobi matatu door handle pushed with 8 N at 0.25 m from the hinge. Cold-call 2 students to substitute values and state units. Connect back to DQB: 'Does this help us begin to explain why the stool fell?'
Lesson 2 **Turning Forces in Action — Calculating and Comparing Moments**	Students use a 30 cm ruler balanced on a pencil pivot, hanging different numbers of identical 10 g masses (or washers) at measured distances from the pivot. They record which side tips and by how much. Teacher circulates and prompts: 'What two things are you changing that affect the tipping?' Allow pairs 2 minutes of unguided exploration before directing structured data collection. Pedagogical note: Insist students	The turning effect of a force is called its MOMENT and is calculated as $M = F \times d$. A heavier load (greater F) or a load placed farther from the pivot (greater d) produces a larger moment and a stronger tendency to rotate. Students should be able to rank a set of force-distance combinations by their moments without a calculator, e.g., 2 N at 0.5 m (1.0 N·m) vs. 3 N at 0.3 m (0.9 N·m). Pedagogical note: Emphasise that moment	Students calculate moments for each mass-distance combination recorded in the observation phase and identify which arrangement produced the greatest turning effect. Teacher models the worked solution on the board, annotating each variable. Connect to the Rift Valley context: a farmer using a long-handled jembe (hoe) exerts a smaller force but achieves the same turning moment as a short-handled tool requiring a

	measure d from the pivot to the point of suspension, not to the end of the ruler.	is a product — doubling d has the same effect as doubling F. This lays the foundation for the Principle of Moments in Lesson 4.	larger force. Cold-call 3 students to explain in their own words why a longer handle makes digging easier, using the word 'moment'. Update the DQB: 'We can now calculate turning effects — what else do we need to explain balance?'
Lesson 3 **Centre of Gravity and the Tipping Point — Why Does the Jiko Stool Fall?**	Students suspend an irregular cardboard shape (cut to resemble the outline of Kenya) from three different points, drawing the plumb-bob vertical each time. They also push a tall, narrow plastic bottle (half-filled with water) and a short, wide mkebe to observe which tips more easily. Teacher asks: 'Where do you think the weight of the whole shape is concentrated?' Allow 1 minute think time, then cold-call 4 students before revealing the centre-of-gravity intersection point.	Every object has a CENTRE OF GRAVITY (CoG) — the single point through which the entire weight of the object appears to act. For a uniform object, the CoG is at the geometric centre; for irregular objects it must be found experimentally (plumb-bob method). A body TOPPLES when a vertical line drawn downward through its CoG falls outside its base of support. Stability increases when the CoG is LOW and the BASE OF SUPPORT is WIDE. Pedagogical note: Students often confuse centre of gravity with centre of mass — clarify that on or near Earth's surface these are effectively the same point.	The jiko stool from Lesson 1 toppled because its narrow three-point base meant that even a small tilt moved the vertical line through its CoG outside the base triangle. Teacher draws a diagram showing the base of support polygon, the CoG position before and after tilting, and the tipping-point condition. Kenyan context examples: a heavily loaded mkokoteni (handcart) with goods stacked high vs. low; a Maasai moran's tall staff balanced upright. Students solve: 'A crate of maize flour 1.2 m tall has its CoG at 0.6 m height and a base width of 0.4 m — at what tilt angle does it topple?' (Answer: $\arctan(0.2/0.6) \approx 18°$.) Connect to DQB: 'We can now predict when an object will topple — how do we calculate exact balance?'
Lesson 4 **The Principle of Moments — Balancing Forces on a Bao Plank**	Students balance a metre ruler (representing a bao game plank) on a central pivot using hanging masses at various positions. They record the load (F), distance from pivot (d), and whether the ruler is balanced. Teacher instructs: 'Record at least five balanced arrangements and five unbalanced ones.' After 10 minutes, pose: 'What mathematical relationship do you notice in your balanced arrangements?' Allow 2 minutes silent analysis before group discussion.	THE PRINCIPLE OF MOMENTS states: for a body in rotational equilibrium, the sum of all CLOCKWISE MOMENTS about any pivot equals the sum of all ANTI-CLOCKWISE MOMENTS about the same pivot. Mathematically: $\Sigma F_{CW} \times d_{CW} = \Sigma F_{ACW} \times d_{ACW}$. This principle allows unknown forces or distances to be calculated. Pedagogical note: Emphasise 'about any pivot' — this is a powerful problem-solving tool. Students should practise choosing a convenient pivot to eliminate unknowns.	Teacher works through a three-load bao-plank problem on the board: a 4 N weight 0.3 m left of pivot and a 2 N weight 0.5 m left must be balanced by a single weight at 0.4 m right — what is that weight? (Solution: CW moment needed = $(4\times0.3)+(2\times0.5) = 1.2+1.0 = 2.2$ N·m; $F = 2.2/0.4 = 5.5$ N.) Cold-call 2 students to verify by substitution. Students then solve: a Kisumu fish-market weighing beam with a 15 N fish 0.2 m from pivot — where must a 5 N counterweight be placed? (Answer: 0.6 m.) Update DQB: 'We can now balance beams — how does this connect to real structures staying stable?'
Lesson 5 **Types of Equilibrium — Stable, Unstable, and Neutral**	Students test three objects: (1) a wide-based clay pot (Kisii soapstone style) tilted slightly and released; (2) a thin marker pen balanced upright on its cap, then nudged; (3) a ball on a flat table, pushed gently. They classify each as returning to original	There are three types of equilibrium: (1) STABLE EQUILIBRIUM — when displaced, the object returns to its original position; its CoG rises upon displacement and the restoring moment acts back. (2) UNSTABLE EQUILIBRIUM — when	Teacher draws three energy-landscape diagrams (valley = stable, hill = unstable, flat = neutral) and maps each Kenyan-context object onto them: the clay pot sits in a valley (stable); the upright pen is on a hilltop (unstable); the ball on a flat table is

	position, toppling further, or moving to a new position. Teacher cold-calls 3 students after 2 minutes: 'Describe exactly what happened and why you think it happened.' Record responses on DQB under the column 'Types of stability'.	displaced, the object moves further away from its original position; its CoG falls upon displacement and the overturning moment grows. (3) NEUTRAL EQUILIBRIUM — when displaced, the object rests in its new position; its CoG neither rises nor falls. Pedagogical note: Link CoG movement to each type explicitly — this is the conceptual core students most often miss.	on a flat plain (neutral). Real-world engineering context: Kenyan SGR train carriages are designed with low CoG and wide wheelbase for stable equilibrium at speed. Students classify five additional objects (a loaded matatu, a Maasai warrior's rungu club, a Mombasa dhow under sail) and justify each classification using CoG and base-of-support language. Update DQB: 'We can classify stability — how do engineers use this to design safe structures?'
Lesson 6 **Applying Moments and Equilibrium — Design, Predict, and Evaluate**	Students examine images and brief case studies: (1) the Thika Superhighway overpass bridge supports; (2) a traditional Luo fishing boat (gwasi) loaded with nets on one side; (3) a construction crane at a Nairobi building site with a counterweight. For each, they identify the pivot, the clockwise and anti-clockwise forces, and predict whether the structure is in equilibrium. Teacher allows 3 minutes group discussion, then cold-calls one student per case to present findings before class.	The Principle of Moments and the concept of centre of gravity together explain and predict the stability of real engineered and natural structures. Engineers deliberately design low centres of gravity, wide bases of support, and balanced moment distributions to ensure stable equilibrium. Moments can be used to calculate unknown support forces in beams, cranes, and bridges. Pedagogical note: This lesson consolidates all prior lessons — use it explicitly as a synthesis. Students should be able to move fluidly between qualitative (type of equilibrium) and quantitative (Principle of Moments calculation) reasoning.	Students apply $M = F \times d$ and the Principle of Moments to solve a capstone problem: a Nairobi construction crane has a 2 000 N load at 8 m from the tower pivot; a counterweight is placed 4 m on the other side — what must the counterweight be for equilibrium? (Answer: 4 000 N.) Teacher then facilitates a full DQB review: students revisit every question posted since Lesson 1 and confirm which are now answered. Final cold-call to 4 students: 'Use today's physics to explain in two sentences why the jiko stool from Lesson 1 toppled.' Pedagogical note: This closing loop to the anchoring phenomenon is essential for consolidating the driving question. Assign a reflection task: students sketch and annotate a stable object from their home, labelling CoG, base of support, and the type of equilibrium.